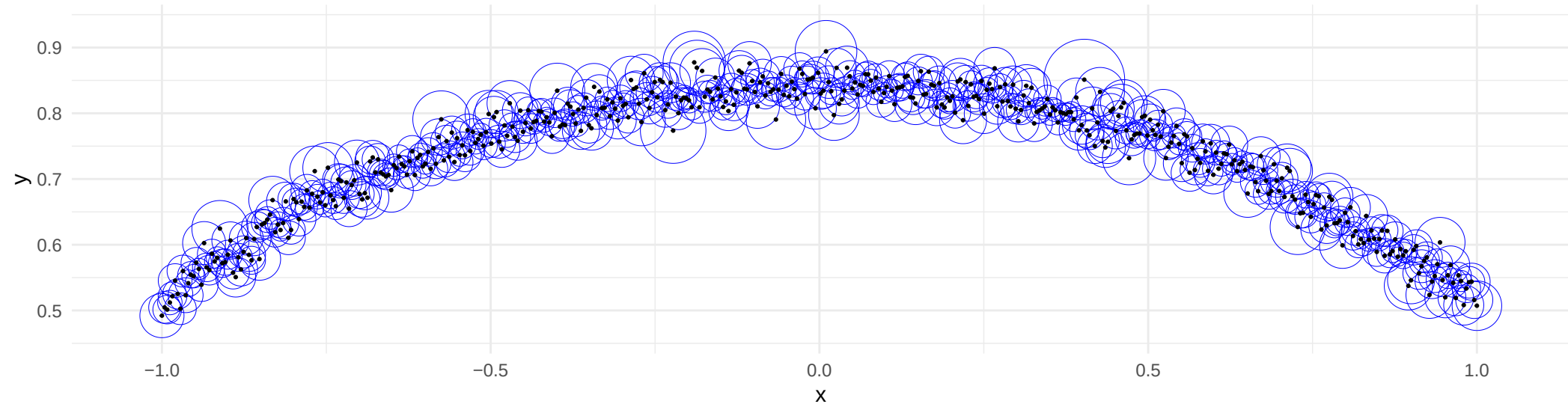
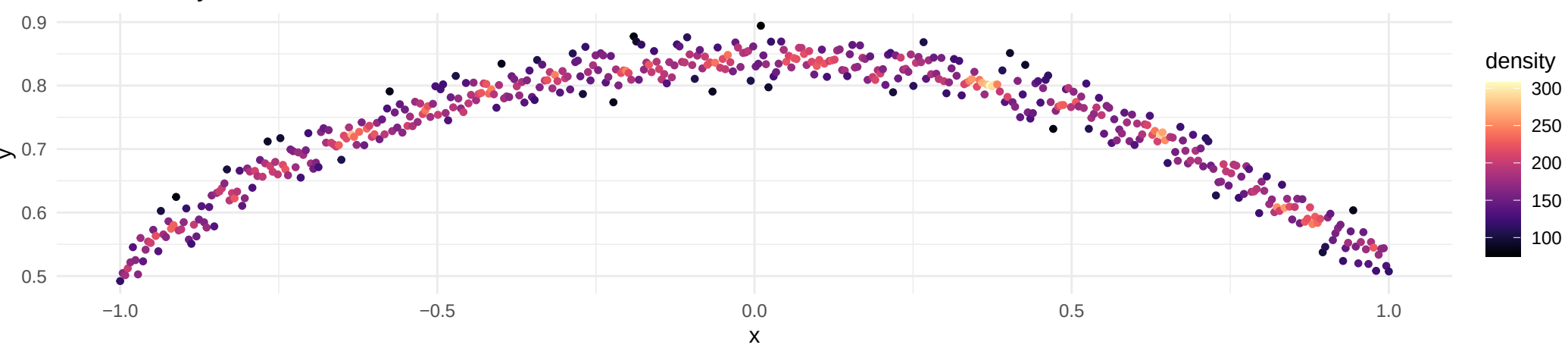


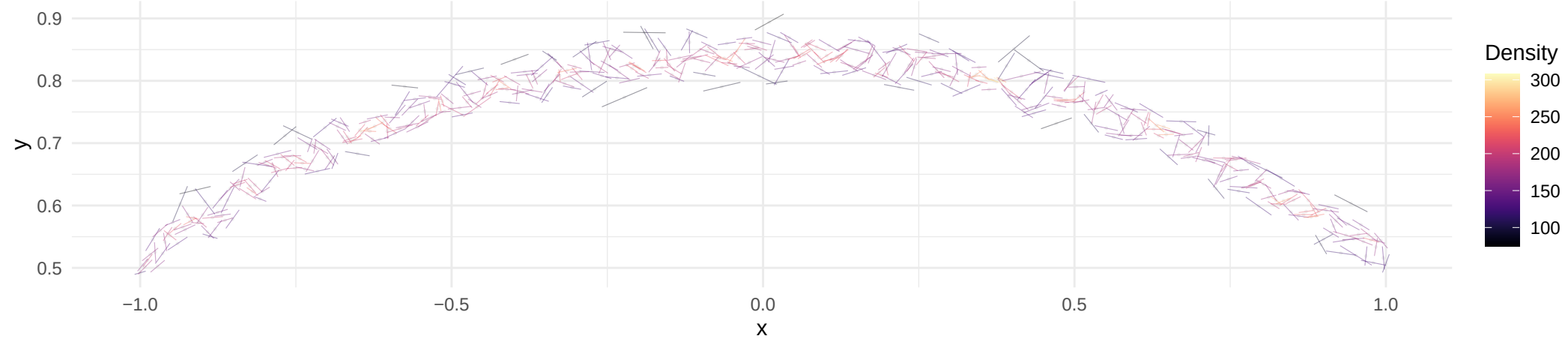
All Adaptive Spheres
Visualizing local neighborhood size (radius) for every point.



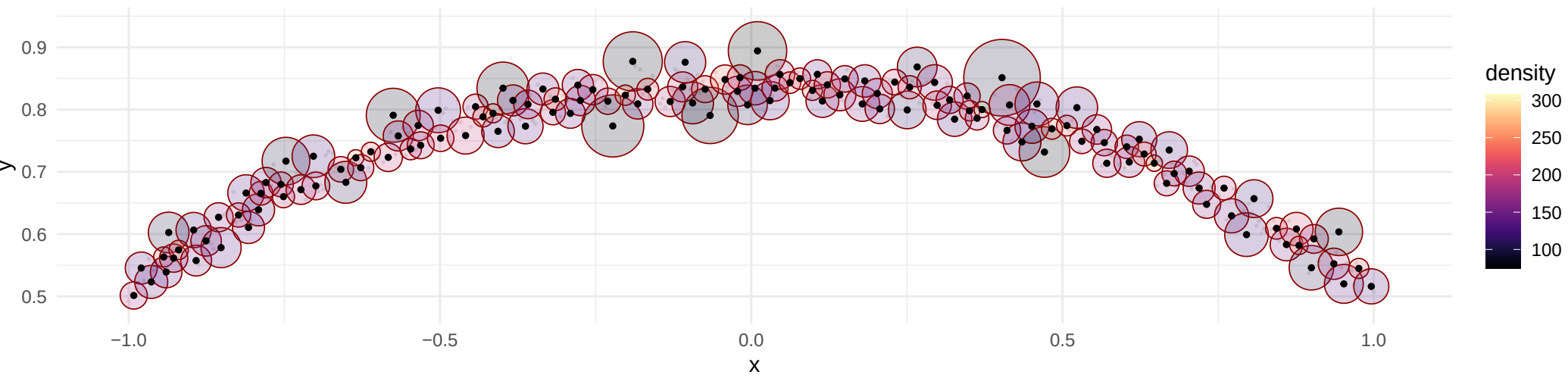
Point Density Distribution



Global Tangent Field (PC1)

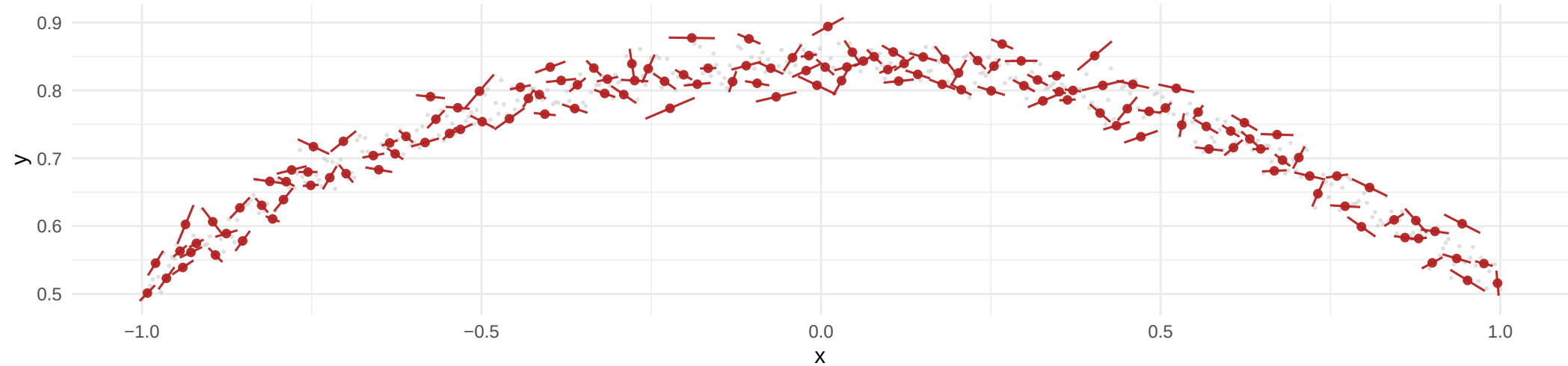


Atlas Selection (Anchors)



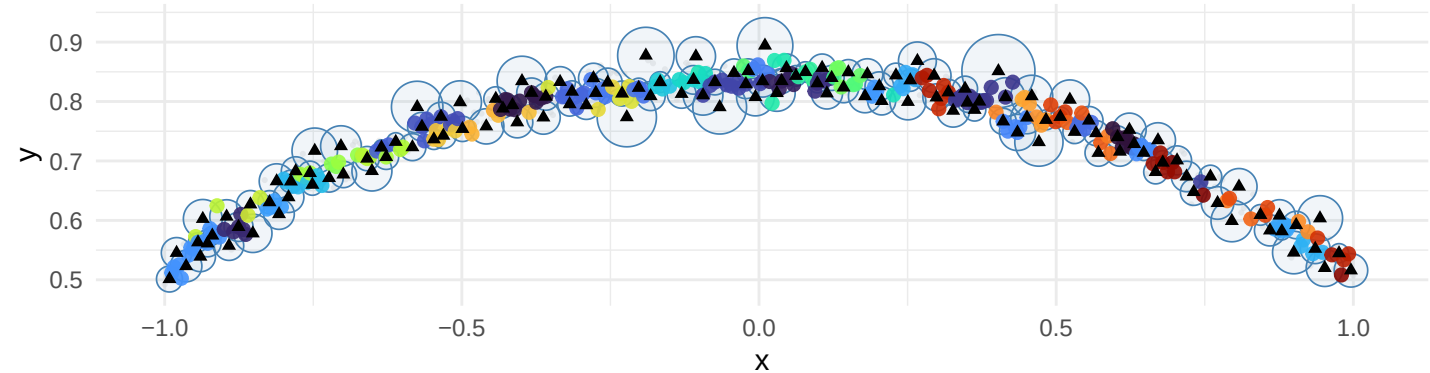
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



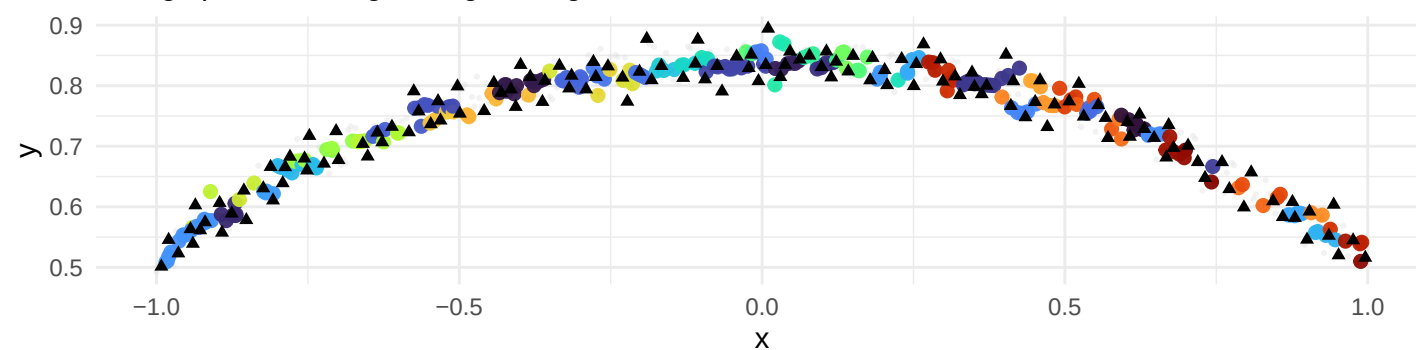
Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01

Region

1	19	37	55	73	91
2	20	38	56	74	92
3	21	39	57	75	93
4	22	40	58	76	94
5	23	41	59	77	95
6	24	42	60	78	96
7	25	43	61	79	97
8	26	44	62	80	98
9	27	45	63	81	99
10	28	46	64	82	100
11	29	47	65	83	101
12	30	48	66	84	102
13	31	49	67	85	103
14	32	50	68	86	
15	33	51	69	87	
16	34	52	70	88	
17	35	53	71	89	
18	36	54	72	90	

Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors

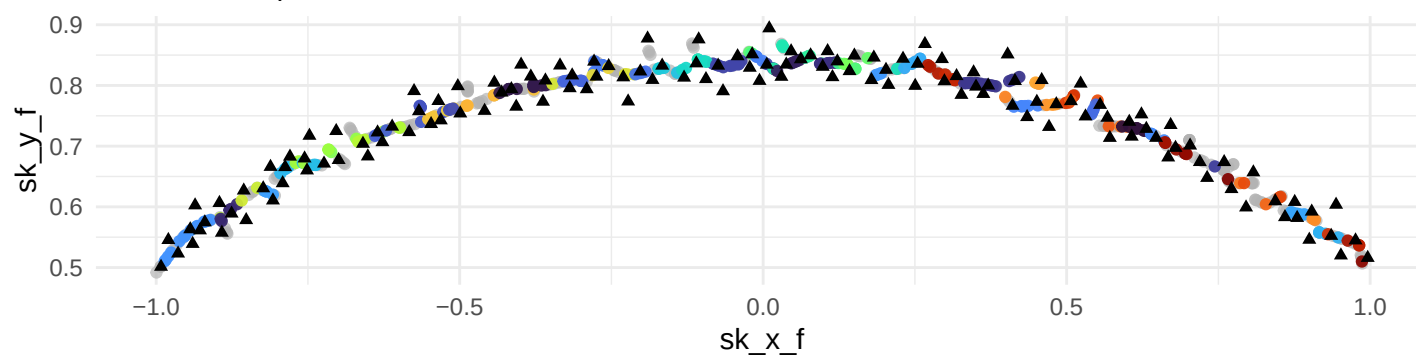


Region

1	19	37	55	73	91
2	20	38	56	74	92
3	21	39	57	75	93
4	22	40	58	76	94
5	23	41	59	77	95
6	24	42	60	78	96
7	25	43	61	79	97
8	26	44	62	80	98
9	27	45	63	81	99
10	28	46	64	82	100
11	29	47	65	83	101
12	30	48	66	84	102
13	31	49	67	85	103
14	32	50	68	86	
15	33	51	69	87	
16	34	52	70	88	
17	35	53	71	89	
18	36	54	72	90	

Final Stitched Backbone (Converged)

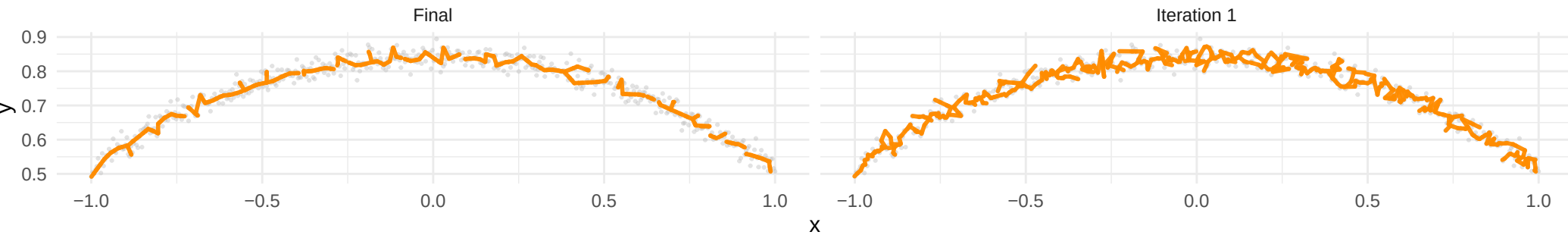
Final stitched points



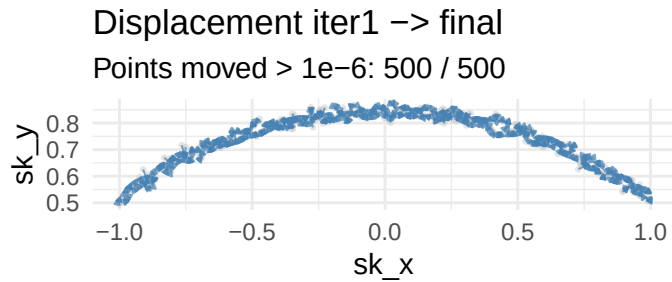
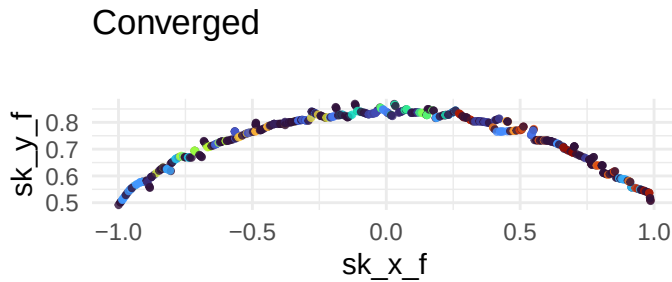
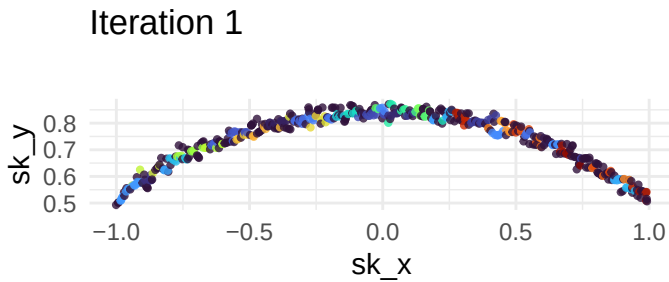
Region

1	19	37	55	73	91
2	20	38	56	74	92
3	21	39	57	75	93
4	22	40	58	76	94
5	23	41	59	77	95
6	24	42	60	78	96
7	25	43	61	79	97
8	26	44	62	80	98
9	27	45	63	81	99
10	28	46	64	82	100
11	29	47	65	83	101
12	30	48	66	84	102
13	31	49	67	85	103
14	32	50	68	86	
15	33	51	69	87	
16	34	52	70	88	
17	35	53	71	89	
18	36	54	72	90	

Backbone Edges



Convergence: iteration 1 vs final



Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01